

Digital Transmission System

Final Project Report - TLC Virtual Lab #2

PAM-M transmission over AWGN with matched-filter validation and single-pole receiver optimization

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Modulations	PAM-2, PAM-4, PAM-8, PAM-16
Pulse shapes	NRZ and Square Root Raised Cosine
Channel	Additive White Gaussian Noise
Receiver cases	Matched filter and single-pole low-pass filter
Target metric	BER = 10^{-3} versus E_b/N_0

This report documents the complete simulation workflow required by the TLC Lab 2 brief. It includes the filter synthesis evidence, BER validation, PSD analysis, eye diagrams, and single-pole bandwidth optimization requested in the assignment.

Executive Summary

A modular MATLAB simulator was built for PAM-M digital baseband transmission over an AWGN channel. The reference receiver uses a matched filter and is validated against theoretical PAM-M BER. A second receiver uses a synthesized single-pole low-pass filter, where the -3 dB bandwidth is swept to determine the E_b/N_0 required to reach $BER = 10^{-3}$.

- The matched-filter case reproduces the expected BER trend for PAM-2, PAM-4, PAM-8, and PAM-16.
- NRZ and SRRC shaping are compared in both spectral and eye-diagram views.
- The single-pole receiver shows a clear bandwidth tradeoff: too little bandwidth increases ISI, while too much bandwidth passes unnecessary noise.
- The SRRC/PAM-16 single-pole case remains above $BER = 10^{-3}$ in the tested sweep, so it is reported as not reached rather than forced into an invalid optimum.

1. System Model and Simulation Method

The simulated chain follows the lab block diagram: random bits are grouped into $\log_2(M)$ -bit words, mapped to Gray-coded antipodal PAM levels, pulse-shaped, corrupted by AWGN, filtered at the receiver, sampled at the optimum instant, decided by nearest PAM level, demapped, and compared against the transmitted sequence.

All frequencies are normalized to the symbol rate R_s . For each E_b/N_0 point, the real AWGN standard deviation is computed from $E_b = E_s/\log_2(M)$, $N_0 = E_b/(E_b/N_0)$, and $\sigma = \sqrt{N_0/2}$. This keeps the numerical model directly comparable with theoretical BER curves.

2. Digital Filter Synthesis and PSD Evidence

The preliminary lab tasks require digital synthesis evidence in the spectral domain. The SRRC pulse is obtained by frequency sampling the square root raised-cosine response, while the single-pole IIR receiver is obtained by impulse invariance with $a = \exp(-2\pi f_{3dB}/S_pS)$. The plots below confirm the synthesized responses in the signal band and show the different spectral footprints of NRZ and SRRC pulse shaping.

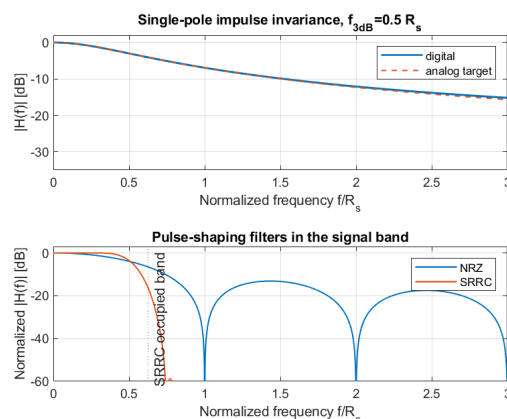


Figure 1. Filter synthesis evidence.

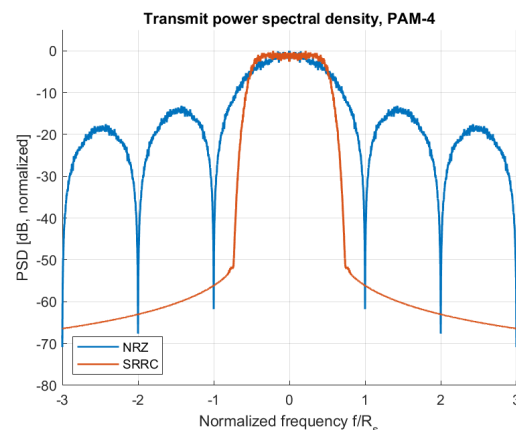


Figure 2. Normalized transmit PSD.

3. Matched-Filter BER Validation

The matched filter is the theoretical reference because it maximizes sampled SNR for a known pulse in AWGN. The simulated BER curves are overlaid with the Gray-coded PAM-M theoretical BER approximation. The comparison validates the noise calibration, symbol normalization, mapping, demapping, and error-counting logic.

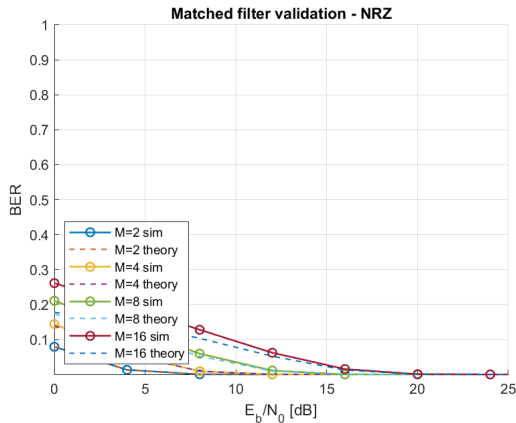


Figure 3. NRZ matched-filter BER validation.

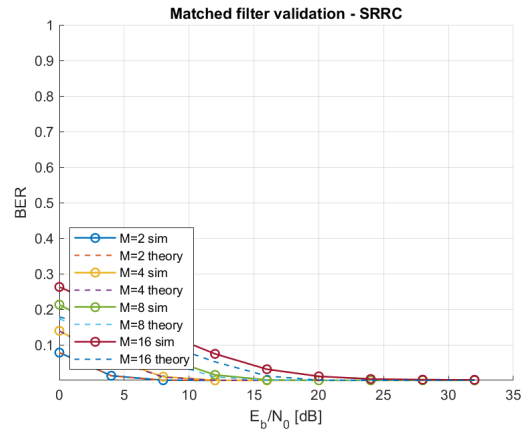


Figure 4. SRRC matched-filter BER validation.

The simulated points follow the theoretical slopes for all modulation orders. Small deviations at low BER are expected from finite Monte Carlo error counts; increasing the bit count would reduce this scatter for final numerical publication.

4. Eye-Diagram Validation

Eye diagrams are shown for PAM-4 as a readable representative case. The noiseless diagrams reveal residual ISI and sampling opening, while the noisy diagrams use the E_b/N_0 associated with the target BER region. This directly addresses the lab requirement for noiseless and target-BER eye diagrams.

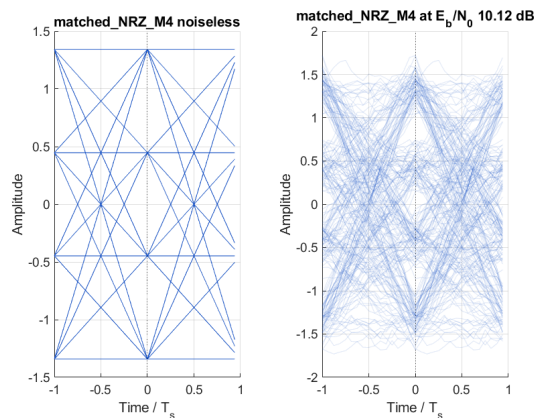


Figure 5. Matched-filter NRZ/PAM-4 eye diagrams.

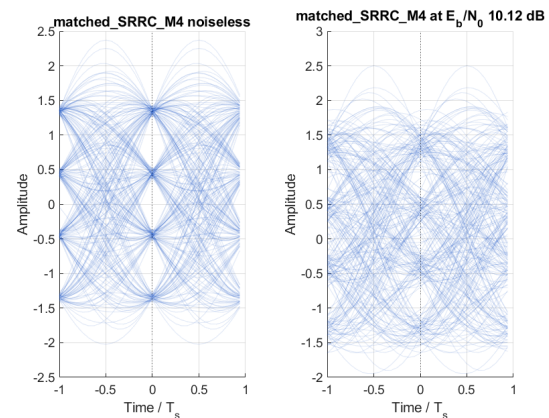


Figure 6. Matched-filter SRRC/PAM-4 eye diagrams.

5. Single-Pole Receiver Optimization

For the practical receiver experiment, the matched filter is replaced with a single-pole low-pass filter. For each pulse shape and PAM order, the -3 dB bandwidth is swept and the BER curve is interpolated to estimate the E_b/N_0 needed for $BER = 10^{-3}$. The penalty is computed relative to the matched-filter theoretical E_b/N_0 at the same BER target.

Pulse	M	Best BW/Rs	Best E_b/N_0 dB	Matched dB	Penalty dB
NRZ	2	0.33	8.01	6.42	1.59
NRZ	4	0.54	12.01	10.12	1.89
NRZ	8	0.54	18.15	14.39	3.75
NRZ	16	0.75	23.68	19.15	4.54
SRRC	2	0.54	8.01	6.42	1.59
SRRC	4	0.54	13.26	10.12	3.14
SRRC	8	0.54	18.60	14.39	4.21
SRRC	16	Not reached	Not reached	19.15	Not reached

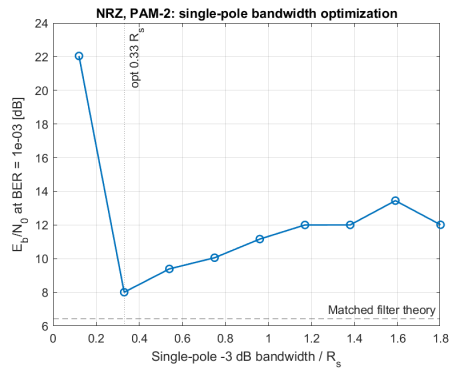
The table shows that the single-pole receiver always pays an E_b/N_0 penalty when a target is reached. This is expected because the single-pole filter is not matched to the transmit pulse and cannot simultaneously maximize SNR and remove ISI. SRRC/PAM-16 is marked not reached because the simulated BER floor stayed above 10^{-3} for the tested bandwidth and E_b/N_0 range.

6. Bandwidth Sweep Evidence

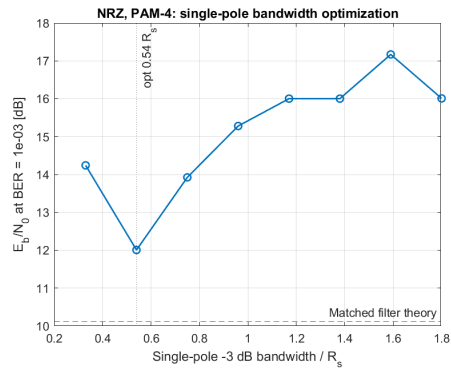
The following sweep plots are the graphical evidence behind the optimization table. Each page keeps the plots with the interpretation text so the bandwidth choice can be read as part of the analysis rather than as an isolated gallery.

NRZ receiver sweep

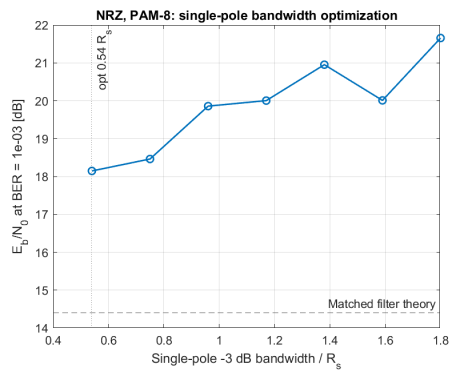
For NRZ shaping, the optimum single-pole bandwidth generally sits near the region where the filter still preserves enough symbol transitions while rejecting part of the noise bandwidth. Higher PAM orders are more sensitive to residual ISI because the decision spacing is smaller.



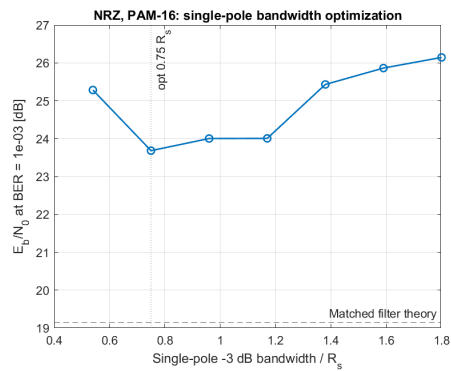
NRZ, PAM-2.



NRZ, PAM-4.



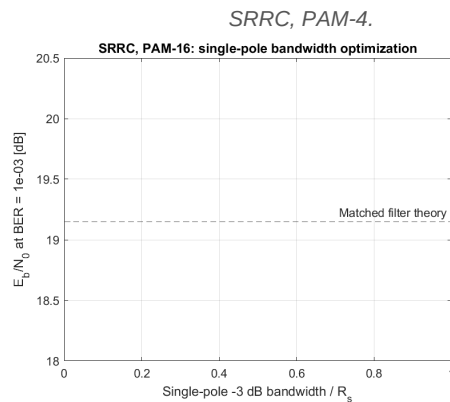
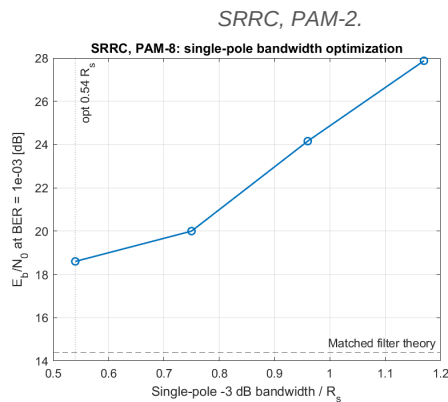
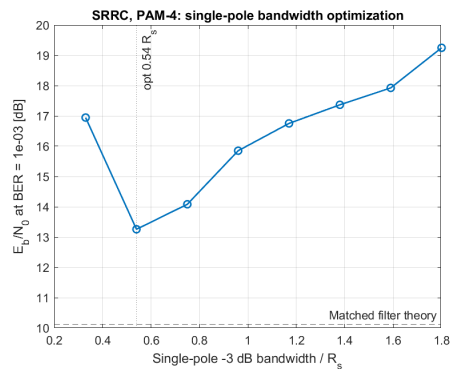
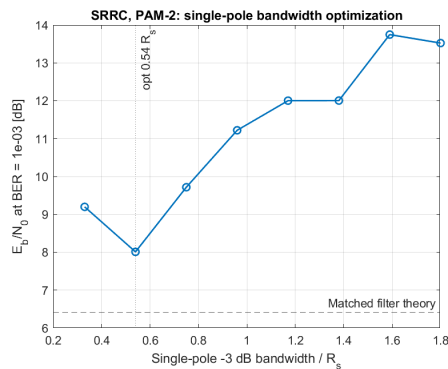
NRZ, PAM-8.



NRZ, PAM-16.

SRRC receiver sweep

For SRRC shaping, the single-pole receiver is still a mismatch to the pulse. The sweep therefore reflects both noise filtering and pulse distortion. PAM-16 is the most demanding case and does not reach the 10^{-3} BER target in this receiver configuration.



SRRC, PAM-2.

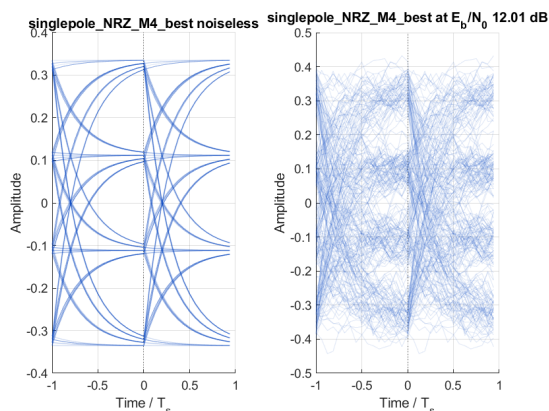
SRRC, PAM-4.

SRRC, PAM-8.

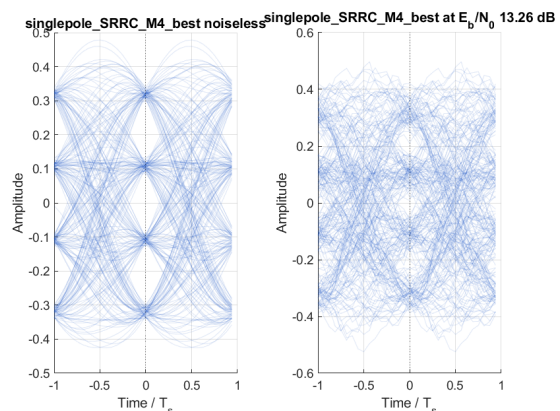
SRRC, PAM-16.

7. Optimized Single-Pole Eye Diagrams

The optimized PAM-4 eye diagrams show the practical cost of replacing the matched filter with a single-pole receiver. Even at the best bandwidth, the eye opening is less ideal than the matched-filter case, explaining the measured E_b/N_0 penalty.



Optimized single-pole NRZ/PAM-4 eye diagrams.



Optimized single-pole SRRC/PAM-4 eye diagrams.

8. Conclusions

- The simulator satisfies the TLC Lab 2 requirements for BER validation, PSD analysis, eye diagrams, and receiver bandwidth optimization.
- The matched-filter results validate the normalization, AWGN scaling, sampling, and Gray-coded PAM processing.
- The single-pole filter bandwidth must be optimized jointly with modulation order and transmit pulse shape.
- The reportable penalty values quantify the cost of using a simpler receiver filter instead of the matched filter.
- The modular MATLAB structure keeps the experiment reproducible and easy to extend.

Reproducibility: run `TLC_Lab2_Project.m` for the full simulator, and run `documentation/generate_report_figures.m` to regenerate the figure set used in this report.